VIKASH KR. GUPTA

Software Developer • *Kolkata, India*

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EDUCATION

St. Thomas' College of Engineering and Technology **2024 – 2028**
Bachelor of Technology in Computer Science & Engineering *CGPA: 7.32 (Till 4th Sem)*

Pearls of God School (ISC) **2022 – 2024**
Class XII (Computer Science) *Percentage: 70.2%*

Pearls of God School (ICSE) **2020 – 2022**
Class X *Percentage: 74.5%*

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript, TypeScript, Shell, Nix
Frameworks & Libraries: React.js, Node.js, Next.js, Three.js, GSAP, Matplotlib, Tkinter
Developer Tools: Git, GitHub, Docker, VS Code, WSL, vim, nano, BlueJ
Databases & Platforms: PostgreSQL, MongoDB, Firebase, GCP

EXPERIENCE

Dexmy Education *Software Engineering Intern*
Full-Stack Development, Backend Engineering & Systems Integration **Aug 2026 – Sep 2026**

- Engineered and integrated core **Dexmy platform features** including a responsive Parent Dashboard with multi-child linking and course discovery, alongside the integration of the existing Node.js/Express Testing service with the FastAPI application, authentication system, role-based authorization, and Test Creator workflows.
- Hardened critical **booking and payment systems** with backend validation, concurrency-safe scheduling, overlap and duplicate protection, teacher availability checks, secure payment verification, webhook idempotency, exactly-once package activation, transaction safety, and automated security/concurrency testing.

PROJECTS

Avataran — The Digital Ramayana Monument *Personal*
Next.js, TypeScript, Tailwind, Node.js, WebSocket, Redis, Framer Motion • *Live Demo*

- Built an interactive digital monument reimagining the Ramayana for modern audiences, replacing dense scripture text with chronological Kanda parallax storytelling, Web Audio sitar/flute orchestration, vertical video reels ("Chitra"), and real-time WebSocket chat ("Samvad").
- Engineered for high concurrency with horizontal WebSocket scaling and a Redis Pub/Sub backplane; shipped as an offline-capable PWA, live on Vercel & Render.

DevTrack *Personal & Hackathon*
React 18, Groq (Llama 3.3), Node.js, Firebase, GitHub API • *Live Demo*

- Built a 'Consistency Engine' merging live GitHub data with learning logs to capture a developer's growth and streaks, not just their current portfolio snapshot.
- Integrated Llama 3.3 for security vulnerability audits and automated codebase complexity hotspot identification; awarded 'Best Innovative Project' for its actionable 'What Next' suggestions and premium skeuomorphic UI.

RailSathi — Smart Railway Commute & Vendor Companion *Personal & Hackathon*
Kotlin, Jetpack Compose, Room SQLite, Node.js, TypeScript, PostgreSQL • *GitHub*

- Built an **offline-first Android platform** connecting suburban railway commuters with verified vendors through coach-aware food ordering, live station awareness, authentic EMU rake visualization, and multilingual accessibility features.
- Engineered a **battery-smart location state machine** that activates GPS only during foreground active travel, with Room-based local persistence, Retrofit networking, and Node.js/Express APIs backed by PostgreSQL.

WORKSHOPS & TRAINING

Kshitij 2026 (IIT Kharagpur) — Summer Web Dev Camp **May 2025 – Jun 2025**
Full-Stack Developer

- Ranked in the Top 10 in IIT Kharagpur's Kshitij Web Development Summer Camp, a competitive remote intensive.
- Applied responsive design and component-based architecture principles under direct mentorship from IIT Kharagpur organizers.

Cloud Study Jam (GDGoC) **2025**

- Completed 2025 Cloud Study Jam syllabus covering cloud storage buckets, virtual machines, and core infra concepts.
- Explored 8+ Google Cloud tools hands-on (Compute Engine, Cloud Storage, Pub/Sub, Gemini API) and shared takeaways through peer sessions.

LEADERSHIP

GDGoC STCET Kolkata **2025 – 2026**
Web & Full-Stack Development Coordinator

- Mentor juniors in web development, Git workflows, and technical problem-solving.
- Manage project workflows and version control using GitHub and agile practices.

ACHIEVEMENTS

- Top 100, Google Solution Challenge 2026
- Winner, CodeFlow SREY 2025
- Finalist, Cosmohacks
- Finalist, CodeFest '26 — IIT BHU
- Top 7, Immpetus: Tech Sprint
- Finalist, FrostHacks S02